A Converse Estimate for the Schlumprecht-Space Szlenk Radius

Automated Math Discovery Smoke Test

June 14, 2026

Source and Open Problem

This packet concerns Proposition 6 and the open problem immediately following it in:

> Tomasz Kochanek and Marek Miarka, *When is the Szlenk derivation of a dual unit ball another ball?*, arXiv:2409.05516 [1].

The source location is page 9 of the paper.

$$\|x_0 + \frac{\varepsilon}{2} e_n\| = \max\left\{r, \frac{\varepsilon}{2}, \frac{r + \frac{\varepsilon}{2}}{\phi(2)}\right\} < 1,$$

which shows that $x_0 \in s_\varepsilon B_{\mathcal{S}}$ and completes the proof. □

Open problem. We do not know if the inequality converse to (6) holds true.

4. Derivations of Baernstein's space

Let $\mathcal{S}$ be Schlumprecht's space [2]. Its norm is given implicitly by

$$\|x\|_{\mathcal{S}} = \max\left\{\|x\|_\infty, \sup_{E_1 < \cdots < E_n} \frac{1}{\varphi(n)} \sum_{i=1}^{n} \|E_i x\|_{\mathcal{S}}\right\}, \qquad \varphi(n) = \log_2(n+1).$$

The case $n = 1$ is redundant, since $\varphi(1) = 1$, so we shall use only $n \geq 2$ in recursive estimates.

Kochanek and Miarka proved

$$r_{\mathcal{S}^*}(\varepsilon) \leq \inf_{n \in \mathbb{N}} \frac{\varphi(n+1) - \varepsilon/2}{\varphi(n)}, \qquad 0 < \varepsilon < 2, \tag{1}$$

and asked whether the converse inequality also holds.

Proof Intuition

The missing direction is a far-coordinate perturbation argument. If a vector $x \in \mathcal{S}$ has norm below the candidate radius, then one wants to add and subtract a sufficiently far basis vector αe_N while staying in the unit ball. The two perturbed vectors converge weakly back to x and remain separated by 2α, which places x in the Szlenk derivation.

The only delicate point is that the scalar majorant must be local: when a restriction F is estimated recursively, the sub-restrictions have to be controlled relative to the majorant of F itself. The needed bound has the form

$$\sum_i \rho(E_i) \leq \varphi(k)\rho(F).$$

The key observation in this revision is that the needed strengthened scalar estimate follows from a simple product inequality for the logarithmic increments of φ.

1 A Logarithmic Product Lemma

Put

$$B_i = \varphi(i+1), \qquad \Delta_i = \varphi(i+1) - \varphi(i) \qquad (i \geq 1).$$

Lemma 1 (monotonicity)**.** *The function*

$$h(u) = \frac{u \log u}{u-1} \qquad (u > 1)$$

is increasing.

Proof. Differentiation gives

$$h'(u) = \frac{u - 1 - \log u}{(u-1)^2} \geq 0,$$

because $\log u \leq u - 1$. □

Lemma 2 (product lemma)**.** *For all $i, j \geq 1$, if $\ell = (i+1)(j+1) - 1$, then*

$$B_\ell \leq B_i B_j, \qquad \Delta_\ell \leq \Delta_i \Delta_j.$$

Proof. It is enough to prove, for $p, q \geq 1$,

$$\log_2(pq+1) \leq \log_2(p+1)\log_2(q+1), \tag{2}$$

$$\log_2\left(1 + \frac{1}{pq}\right) \leq \log_2\left(1 + \frac{1}{p}\right)\log_2\left(1 + \frac{1}{q}\right). \tag{3}$$

For fixed $p \geq 1$, define

$$F_p(q) = \frac{\log(pq+1)}{\log(q+1)} \qquad (q > 1),$$

where log is the natural logarithm. The derivative has the sign of

$$p(q+1)\log(q+1) - (pq+1)\log(pq+1).$$

With $u = q+1$ and $v = pq+1 = 1 + p(u-1)$, this sign is non-positive by Lemma 1. Indeed, $v \geq u$, and hence

$$\frac{v \log v}{v-1} \geq \frac{u \log u}{u-1}.$$

Since $v - 1 = p(u-1)$, this gives $v \log v \geq pu \log u$, which is exactly the required non-positivity of the derivative sign. Hence F_p is decreasing, and $F_p(q) \leq F_p(1) = \log(p+1)/\log 2$. This is (2).

For (3), write $a = 1/p$ and $b = 1/q$. For fixed $0 < a \leq 1$, define

$$G_a(b) = \frac{\log(1+ab)}{\log(1+b)} \qquad (0 < b \leq 1).$$

The derivative has the sign of

$$a(1+b)\log(1+b) - (1+ab)\log(1+ab).$$

With $u = 1+b$ and $v = 1+ab = 1 + a(u-1)$, this sign is non-negative by Lemma 1. Here $v \leq u$, and so

$$\frac{v \log v}{v-1} \leq \frac{u \log u}{u-1}.$$

Since $v - 1 = a(u-1)$, this gives $v \log v \leq au \log u$, which is exactly the required non-negativity of the derivative sign. Thus G_a is increasing on $(0, 1]$, and $G_a(b) \leq G_a(1) = \log(1+a)/\log 2$. This is (3).

Taking $p = i+1$ and $q = j+1$ gives the asserted inequalities for B_ℓ and Δ_ℓ. □

2 The Strengthened Scalar Estimate

For $0 < \alpha < 1$, set

$$a_\alpha = \inf_{n\in\mathbb{N}} \frac{\varphi(n+1)-\alpha}{\varphi(n)}.$$

Then $0 < a_\alpha < 1$. Positivity follows from $\varphi(n+1) > 1 > \alpha$ for every $n \geq 1$. For the upper bound, use $\varphi(n+1) - \varphi(n) \to 0$: choose n with $\varphi(n+1) - \varphi(n) < \alpha$. Then

$$\frac{\varphi(n+1)-\alpha}{\varphi(n)} < 1,$$

and hence $a_\alpha < 1$. Equivalently,

$$\varphi(n)a_\alpha + \alpha \leq \varphi(n+1) \qquad (n \in \mathbb{N}). \tag{4}$$

For $u \geq 0$, define

$$M_\alpha(u) = \max\left\{u, \alpha, \sup_{m\in\mathbb{N}} \frac{\varphi(m)u+\alpha}{\varphi(m+1)}\right\}.$$

If $0 \leq u < a_\alpha$, then $M_\alpha(u) \leq 1$: indeed $u < 1$ and $\alpha < 1$, while (4) gives

$$\frac{\varphi(m)u+\alpha}{\varphi(m+1)} \leq \frac{\varphi(m)a_\alpha+\alpha}{\varphi(m+1)} \leq 1 \qquad (m \in \mathbb{N}).$$

Lemma 3 (strengthened scalar estimate)**.** *Let $0 < \alpha < 1$, $0 \leq s \leq t$, and $k \geq 2$. Then*

$$M_\alpha(s) + \min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k)M_\alpha(t). \tag{5}$$

Proof. Write

$$a = \varphi(k-1), \qquad b = \varphi(k), \qquad \delta = b - a.$$

Since

$$\delta = \log_2(k+1) - \log_2(k) = \log_2\left(1 + \frac{1}{k}\right),$$

we have $0 < \delta < 1$, and therefore $\delta t \in [0, t]$. It suffices to prove (5) for each affine term in the maximum defining $M_\alpha(s)$.

For the term s, use the second member of the minimum:

$$s + \min\{at, bt - s\} \leq bt \leq bM_\alpha(t).$$

For the term α, use the first member of the minimum and the $(k-1)$-st affine term in $M_\alpha(t)$:

$$\alpha + \min\{at, bt - s\} \leq \alpha + at \leq bM_\alpha(t).$$

It remains to consider the m-th affine term. Put

$$c = \varphi(m), \qquad d = \varphi(m+1), \qquad \eta = d - c.$$

For fixed t, the function

$$s \mapsto \frac{cs+\alpha}{d} + \min\{at, bt - s\}$$

increases on $[0, \delta t]$ and decreases on $[\delta t, t]$. Its maximum over $0 \le s \le t$ is therefore attained at $s = \delta t$, where it equals

$$\frac{\alpha + t(c\delta + ad)}{d} = \frac{\alpha + t(bd - \delta\eta)}{d}.$$

Thus it is enough to prove

$$M_\alpha(t) \ge \frac{\alpha + t(bd - \delta\eta)}{bd}. \tag{6}$$

Apply Lemma 2 with $i = k - 1$ and $j = m$. There is $\ell \in \mathbb{N}$ such that, with

$$D = B_\ell = \varphi(\ell + 1), \qquad \lambda = \Delta_\ell = \varphi(\ell + 1) - \varphi(\ell),$$

we have $D \le bd$ and $\lambda \le \delta\eta$. The ℓ-th affine term in $M_\alpha(t)$ gives

$$M_\alpha(t) \ge \frac{\varphi(\ell)t + \alpha}{\varphi(\ell + 1)} = t + \frac{\alpha - \lambda t}{D}.$$

If $\alpha - \delta\eta t < 0$, the right-hand side of (6) is below t, while $M_\alpha(t) \ge t$. If $\alpha - \delta\eta t \ge 0$, then $\alpha - \lambda t \ge \alpha - \delta\eta t \ge 0$ and $D \le bd$, so

$$M_\alpha(t) \ge t + \frac{\alpha - \delta\eta t}{bd} = \frac{\alpha + t(bd - \delta\eta)}{bd}.$$

This proves (6), hence (5). □

Remark 4. *The factor $M_\alpha(t)$ on the right-hand side of Lemma 3 is what makes the estimate stable under the recursive definition of the Schlumprecht norm.*

3 The Far-Coordinate Perturbation

Define the iterative norms by

$$\|x\|_0 = \|x\|_\infty, \qquad \|x\|_{r+1} = \max\left\{\|x\|_\infty, \sup_{k \ge 2} \sup_{E_1 < \cdots < E_k} \frac{1}{\varphi(k)} \sum_{i=1}^{k} \|E_i x\|_r\right\}.$$

Then $\|x\|_{\mathcal{S}} = \sup_r \|x\|_r$.

Lemma 5 (comparison by local majorants)**.** *Let $y \in c_{00}$. Suppose that to every finite $F \subset \mathbb{N}$ one assigns $\rho(F) \ge 0$ such that*

$$\begin{aligned} \|Fy\|_\infty &\le \rho(F), \\ G \subset F &\Longrightarrow \rho(G) \le \rho(F), \\ \sum_{i=1}^{k} \rho(E_i) &\le \varphi(k)\rho(F) \end{aligned}$$

whenever $k \ge 2$ and $E_1 < \cdots < E_k$ are finite subsets of F. Then

$$\|Fy\|_{\mathcal{S}} \le \rho(F) \qquad (F \subset \mathbb{N} \textit{ finite}).$$

Proof. Induct on the iterative norms. The assertion is true for $\|\cdot\|_0$ by the first assumption. Suppose it is true for $\|\cdot\|_r$. Fix F and an admissible family $E_1 < \cdots < E_k$, $k \geq 2$. Put $G_i = E_i \cap F$ and discard empty intersections.

If no G_i remains, the sum is zero. If exactly one set G remains, then

$$\|Gy\|_r \leq \rho(G) \leq \rho(F) \leq \varphi(k)\rho(F).$$

The last inequality uses $k \geq 2$, hence $\varphi(k) \geq 1$. If $q \geq 2$ sets $G_1 < \cdots < G_q$ remain, then $q \leq k$, and

$$\sum_{j=1}^{q} \|G_j y\|_r \leq \sum_{j=1}^{q} \rho(G_j) \leq \varphi(q)\rho(F) \leq \varphi(k)\rho(F).$$

Thus every admissible sum in the definition of $\|Fy\|_{r+1}$ is bounded by $\varphi(k)\rho(F)$, and so $\|Fy\|_{r+1} \leq \rho(F)$. Taking the supremum over r proves the claim. $\square$

Lemma 6 (far-coordinate unit-ball lemma)**.** *Let $0 < \alpha < 1$. If $x \in c_{00}$ and $\|x\|_{\mathcal{S}} < a_\alpha$, then, for every $N > \max \operatorname{supp} x$,*

$$\|x + \alpha e_N\|_{\mathcal{S}} \leq 1, \qquad \|x - \alpha e_N\|_{\mathcal{S}} \leq 1.$$

Proof. By 1-unconditionality and $N \notin \operatorname{supp} x$,

$$\|x + \alpha e_N\|_{\mathcal{S}} = \||x| + \alpha e_N\|_{\mathcal{S}}, \qquad \|x - \alpha e_N\|_{\mathcal{S}} = \||x| + \alpha e_N\|_{\mathcal{S}}.$$

Thus it is enough to consider $x \geq 0$ and $y = x + \alpha e_N$. For each finite $F \subset \mathbb{N}$, define

$$\rho(F) = \begin{cases} \|Fx\|_{\mathcal{S}}, & N \notin F, \\ M_\alpha(\|Fx\|_{\mathcal{S}}), & N \in F. \end{cases}$$

The lattice monotonicity of the Schlumprecht norm and the monotonicity of M_α give $G \subset F \Rightarrow \rho(G) \leq \rho(F)$. Also $\|Fy\|_\infty \leq \rho(F)$, since $M_\alpha(u) \geq u$ and $M_\alpha(u) \geq \alpha$.

We verify the local majorant inequality. Fix finite F and $E_1 < \cdots < E_k \subset F$, $k \geq 2$. If $N \notin F$, then

$$\sum_i \rho(E_i) = \sum_i \|E_i x\|_{\mathcal{S}} \leq \varphi(k)\|Fx\|_{\mathcal{S}} = \varphi(k)\rho(F).$$

Assume $N \in F$ and set $t = \|Fx\|_{\mathcal{S}}$. If none of the E_i contains N, then

$$\sum_i \rho(E_i) = \sum_i \|E_i x\|_{\mathcal{S}} \leq \varphi(k)t \leq \varphi(k)M_\alpha(t) = \varphi(k)\rho(F).$$

If exactly one set, say E_j, contains N, put $s = \|E_j x\|_{\mathcal{S}}$. The remaining pieces satisfy

$$\sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \varphi(k-1)t, \qquad \sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \varphi(k)t - s.$$

For the first inequality, if $k = 2$ then there is only one remaining piece, say E_i, and

$$\sum_{i \neq j} \|E_i x\|_{\mathcal{S}} = \|E_i x\|_{\mathcal{S}} \leq \|Fx\|_{\mathcal{S}} = t = \varphi(1)t.$$

If $k > 2$, the $k-1$ remaining pieces form an ordered admissible family, so the norm recursion gives the bound with $\varphi(k-1)$. The second inequality follows from the full k-tuple. Lemma 3 gives

$$\sum_i \rho(E_i) \leq M_\alpha(s) + \min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k)M_\alpha(t) = \varphi(k)\rho(F).$$

Lemma 5 therefore yields $\|Fy\|_{\mathcal{S}} \leq \rho(F)$ for every finite F. Taking $F = \operatorname{supp} x \cup \{N\}$ gives

$$\|x + \alpha e_N\|_{\mathcal{S}} \leq M_\alpha(\|x\|_{\mathcal{S}}) \leq 1,$$

because $\|x\|_{\mathcal{S}} < a_\alpha$. The estimate for $x - \alpha e_N$ follows from the unconditionality reduction at the start of the proof. □

4 The Converse Formula

For a Banach space X, use the source paper's notation

$$r_X(\varepsilon) = \sup\{r > 0 : rB_{X^*} \subset s_\varepsilon(B_{X^*})\}.$$

For $X = \mathcal{S}^*$, identify $X^* = \mathcal{S}^{**}$ with $\mathcal{S}$, since $\mathcal{S}$ is reflexive.

Theorem 7. *For every* $0 < \varepsilon < 2$*,*

$$r_{\mathcal{S}^*}(\varepsilon) = \inf_{n\in\mathbb{N}} \frac{\log_2(n+2) - \varepsilon/2}{\log_2(n+1)}.$$

Proof. Let $\beta = \varepsilon/2$ and

$$a_\beta = \inf_{n\in\mathbb{N}} \frac{\varphi(n+1) - \beta}{\varphi(n)}.$$

The upper bound $r_{\mathcal{S}^*}(\varepsilon) \leq a_\beta$ is (1), proved by Kochanek and Miarka. We prove the reverse inequality.

First note that $\alpha \mapsto a_\alpha$ is continuous on $(0, 1)$: it is the infimum of affine functions with slopes bounded in absolute value by 1. Let $x \in c_{00}$ and $\|x\|_{\mathcal{S}} < a_\beta$. Choose $\alpha \in (\beta, 1)$ close enough to β that $\|x\|_{\mathcal{S}} < a_\alpha$. For every $N > \max \operatorname{supp} x$, Lemma 6 gives

$$x + \alpha e_N,\ x - \alpha e_N \in B_{\mathcal{S}}.$$

The basis of $\mathcal{S}$ is weakly null, because $\mathcal{S}$ is reflexive and its basis is shrinking. Hence both sequences $x + \alpha e_N$ and $x - \alpha e_N$ converge weakly, equivalently weak-star in $\mathcal{S} = (\mathcal{S}^*)^*$, to x. Their mutual distance is

$$\|(x + \alpha e_N) - (x - \alpha e_N)\|_{\mathcal{S}} = 2\alpha > \varepsilon.$$

Thus every weak-star neighborhood of x intersects $B_{\mathcal{S}}$ in a set of diameter greater than ε, and $x \in s_\varepsilon(B_{\mathcal{S}})$.

Now fix $0 < r < a_\beta$ and $z \in rB_{\mathcal{S}}$. Choose $z_m \in c_{00}$ with $z_m \to z$ in norm. For all large m, $\|z_m\|_{\mathcal{S}} < a_\beta$, hence $z_m \in s_\varepsilon(B_{\mathcal{S}})$. The Szlenk derivation of the weak-star compact ball is weak-star closed, hence norm closed. Indeed, if K is weak-star compact and $x \in K \setminus s_\varepsilon(K)$, then some weak-star open neighborhood $V \ni x$ has $\operatorname{diam}(K \cap V) \leq \varepsilon$; the same V shows every $y \in K \cap V$ also lies outside $s_\varepsilon(K)$, so $K \setminus s_\varepsilon(K)$ is relatively weak-star open. Therefore $z \in s_\varepsilon(B_{\mathcal{S}})$. Since this holds for every $z \in rB_{\mathcal{S}}$ and every $r < a_\beta$, we have $r_{\mathcal{S}^*}(\varepsilon) \geq a_\beta$. Combined with the source paper's upper estimate, this proves the formula. □

Verification Notes

The packet includes `code/check_scalar_and_finite_cases.py`. The scalar estimate is proved by Lemma 2 and Lemma 3; the code is only a sanity check. It tests the scalar inequalities on a large finite grid and random sample and computes representative finite-support Schlumprecht norms by interval dynamic programming. The command used was:

```
conda run --no-capture-output -n sandbox python
code/check_scalar_and_finite_cases.py
```

Novelty and Literature Check

Bounded check performed on June 14, 2026: searched the arXiv page for arXiv:2409.05516 and web queries for the exact source title together with "Schlumprecht", "Szlenk", "converse", and "r_{S^*}". This found the source paper and no later paper explicitly resolving the converse formula. This is not an exhaustive bibliographic search, but no immediate literature-already-answered match was found.

Human Review Recommendation

This should be reviewed as a candidate full solution. The most important points to check are the product lemma, the strengthened scalar estimate, and the local majorant comparison lemma.

References

[1] T. Kochanek and M. Miarka, *When is the Szlenk derivation of a dual unit ball another ball?*, arXiv:2409.05516 (2024).

[2] T. Schlumprecht, *An arbitrarily distortable Banach space*, Israel J. Math. 76 (1991), 81–95.

Self-Contained Better-Notation Rewrite of the Schlumprecht Szlenk-Radius Converse

Human-requested companion note

Important note. This document is not the original LLM-generated output. It is a human-requested rewrite, requested by Antonio Acuaviva, of the proposed argument with improved notation and with the details made self-contained. The mathematical content is intended to be the same as in the original solution packet, but the notation has been reorganized to make the proof easier to check.

1 Setup

Let

$$\varphi(n) = \log_2(n+1) \qquad (n \in \mathbb{N}).$$

Schlumprecht's space $\mathcal{S}$ is the completion of c_{00} under the norm

$$\|x\|_{\mathcal{S}} = \max\left\{\|x\|_\infty, \sup_{n\geq 2,\ E_1<\cdots<E_n} \frac{1}{\varphi(n)}\sum_{i=1}^n \|E_i x\|_{\mathcal{S}}\right\}.$$

Here Ex denotes the coordinate restriction of x to the finite set E, and $E_1 < \cdots < E_n$ means the sets are successive.

We use the standard facts that the canonical basis of $\mathcal{S}$ is 1-unconditional and shrinking, and that $\mathcal{S}$ is reflexive. Thus $e_N \to 0$ weakly in $\mathcal{S}$.

For a weak-star compact set $K \subset X^*$, the Szlenk derivation is

$$s_\varepsilon(K) = \{x^* \in K : \operatorname{diam}(K \cap V) > \varepsilon \text{ for every weak-star neighbourhood } V \ni x^*\}.$$

In the present case $X = \mathcal{S}^*$, so $X^* = \mathcal{S}$. The corresponding inner Szlenk radius is

$$r_{\mathcal{S}^*}(\varepsilon) = \sup\{r > 0 : rB_{\mathcal{S}} \subset s_\varepsilon(B_{\mathcal{S}})\}.$$

For $0 < \alpha < 1$, define

$$a_\alpha = \inf_{n\in\mathbb{N}} \frac{\varphi(n+1) - \alpha}{\varphi(n)}.$$

The known upper bound from the source paper is

$$r_{\mathcal{S}^*}(\varepsilon) \leq a_{\varepsilon/2}.$$

The goal of the proposed argument is to prove the reverse inequality.

Theorem. *For* $0 < \varepsilon < 2$*,*

$$r_{\mathcal{S}^*}(\varepsilon) = \inf_{n\in\mathbb{N}} \frac{\varphi(n+1) - \varepsilon/2}{\varphi(n)}.$$

2 Logarithmic notation

For $i \geq 1$, write

$$B_i = \varphi(i+1), \qquad \Delta_i = \varphi(i+1) - \varphi(i).$$

Thus

$$B_i = \log_2(i+2), \qquad \Delta_i = \log_2\left(1 + \frac{1}{i+1}\right).$$

This notation is kept throughout the proof. In particular, when the product lemma is applied later with $i = k-1$ and $j = m$, the relevant quantities remain visibly

$$\Delta_{k-1}, \qquad \Delta_m, \qquad B_\ell, \qquad \Delta_\ell.$$

3 The product lemma

We first record the elementary monotonicity fact behind the product lemma.

Lemma. *The function*

$$h(u) = \frac{u \log u}{u-1} \qquad (u > 1)$$

is increasing.

Proof. Differentiate:

$$h'(u) = \frac{(u-1)(\log u + 1) - u \log u}{(u-1)^2} = \frac{u - 1 - \log u}{(u-1)^2}.$$

Since $\log u \leq u - 1$, we have $h'(u) \geq 0$. $\square$

Lemma (Product lemma)**.** *For $i, j \geq 1$, put*

$$\ell = (i+1)(j+1) - 1.$$

Then

$$B_\ell \leq B_i B_j, \qquad \Delta_\ell \leq \Delta_i \Delta_j.$$

Proof. The first inequality is equivalent to

$$\log_2(pq+1) \leq \log_2(p+1)\log_2(q+1) \qquad (p, q \geq 2),$$

where $p = i+1$ and $q = j+1$. For fixed p, consider

$$F_p(q) = \frac{\log(pq+1)}{\log(q+1)}.$$

A quotient-rule calculation gives

$$F_p'(q) = \frac{p(q+1)\log(q+1) - (pq+1)\log(pq+1)}{(pq+1)(q+1)(\log(q+1))^2}.$$

The numerator is non-positive because it is the comparison

$$\frac{(q+1)\log(q+1)}{q} \leq \frac{(pq+1)\log(pq+1)}{pq},$$

which follows from the monotonicity of h. Hence F_p is decreasing, so $F_p(q) \leq F_p(1)$. This gives

$$\frac{\log(pq+1)}{\log(q+1)} \leq \frac{\log(p+1)}{\log 2},$$

which is the first inequality.

For the increment inequality, put $a = 1/p$ and $b = 1/q$. It is enough to prove

$$\log_2(1+ab) \le \log_2(1+a)\log_2(1+b) \qquad (0 < a, b \le 1).$$

For fixed a, define

$$G_a(b) = \frac{\log(1+ab)}{\log(1+b)}.$$

Again differentiating,

$$G_a'(b) = \frac{a(1+b)\log(1+b) - (1+ab)\log(1+ab)}{(1+ab)(1+b)(\log(1+b))^2}.$$

Let $u = 1+b$ and $v = 1+ab$. Since $0 < a \le 1$, $v \le u$, and

$$a(u-1) = v-1.$$

The numerator is

$$au\log u - v\log v = (v-1)\left(\frac{u\log u}{u-1} - \frac{v\log v}{v-1}\right) \ge 0$$

by the monotonicity of h. Hence G_a is increasing, so

$$G_a(b) \le G_a(1) = \frac{\log(1+a)}{\log 2}.$$

This proves the increment inequality. □

4 The scalar majorant

First note that $0 < a_\alpha < 1$. For positivity, since $\varphi(n+1) > \varphi(n)$ and $\varphi(n) \ge 1$,

$$\frac{\varphi(n+1)-\alpha}{\varphi(n)} > \frac{\varphi(n)-\alpha}{\varphi(n)} = 1 - \frac{\alpha}{\varphi(n)} \ge 1-\alpha > 0.$$

Thus $a_\alpha \ge 1-\alpha > 0$. For the upper bound, use $\varphi(n+1)-\varphi(n) \to 0$. Choose n with

$$\varphi(n+1) - \varphi(n) < \alpha.$$

Then

$$\frac{\varphi(n+1)-\alpha}{\varphi(n)} < 1,$$

so $a_\alpha < 1$.

From the definition of the infimum, for every $n \in \mathbb{N}$,

$$a_\alpha \le \frac{\varphi(n+1)-\alpha}{\varphi(n)}.$$

Equivalently,

$$\varphi(n)a_\alpha + \alpha \le \varphi(n+1).$$

For $u \ge 0$, define

$$M_\alpha(u) = \max\left\{u, \alpha, \sup_{m\in\mathbb{N}} \frac{\varphi(m)u+\alpha}{\varphi(m+1)}\right\}.$$

This scalar function is designed to dominate the cost of adding one far coordinate of size α.

If $0 \le u < a_\alpha$, then $M_\alpha(u) \le 1$. Indeed, $u < 1$ and $\alpha < 1$, while for every m,

$$\frac{\varphi(m)u+\alpha}{\varphi(m+1)} \le \frac{\varphi(m)a_\alpha+\alpha}{\varphi(m+1)} \le 1.$$

5 Strengthened scalar estimate

Lemma (Strengthened scalar estimate)**.** *Let $0 < \alpha < 1$, $0 \le s \le t$, and $k \ge 2$. Then*

$$M_\alpha(s) + \min\{\varphi(k-1)t, \varphi(k)t - s\} \le \varphi(k)M_\alpha(t).$$

Proof. We prove the estimate separately for each type of term appearing in $M_\alpha(s)$.

First consider the term s. Since

$$\min\{\varphi(k-1)t, \varphi(k)t - s\} \le \varphi(k)t - s,$$

we have

$$s + \min\{\varphi(k-1)t, \varphi(k)t - s\} \le \varphi(k)t \le \varphi(k)M_\alpha(t).$$

Next consider the term α. Since

$$\min\{\varphi(k-1)t, \varphi(k)t - s\} \le \varphi(k-1)t,$$

we get

$$\alpha + \min\{\varphi(k-1)t, \varphi(k)t - s\} \le \alpha + \varphi(k-1)t.$$

But $M_\alpha(t)$ contains the affine term indexed by $k-1$:

$$M_\alpha(t) \ge \frac{\varphi(k-1)t + \alpha}{\varphi(k)}.$$

Thus

$$\alpha + \varphi(k-1)t \le \varphi(k)M_\alpha(t).$$

It remains to consider an affine term indexed by m:

$$\frac{\varphi(m)s + \alpha}{\varphi(m+1)}.$$

For fixed t, put

$$F_m(s) = \frac{\varphi(m)s + \alpha}{\varphi(m+1)} + \min\{\varphi(k-1)t, \varphi(k)t - s\}.$$

The two quantities inside the minimum are equal when

$$\varphi(k-1)t = \varphi(k)t - s,$$

that is, when

$$s = (\varphi(k) - \varphi(k-1))t = \Delta_{k-1}t.$$

On $0 \le s \le \Delta_{k-1}t$, the minimum is $\varphi(k-1)t$, so F_m is increasing. On $\Delta_{k-1}t \le s \le t$, the minimum is $\varphi(k)t - s$, and the slope of F_m is

$$\frac{\varphi(m)}{\varphi(m+1)} - 1 < 0.$$

Hence F_m is maximized at $s = \Delta_{k-1}t$.

At that point,

$$F_m(\Delta_{k-1}t) = \frac{\varphi(m)\Delta_{k-1}t + \alpha}{\varphi(m+1)} + \varphi(k-1)t.$$

Putting this over the denominator $\varphi(m+1)$, we get

$$F_m(\Delta_{k-1}t) = \frac{\alpha + t(\varphi(m)\Delta_{k-1} + \varphi(k-1)\varphi(m+1))}{\varphi(m+1)}.$$

Using

$$\varphi(k) = \varphi(k-1) + \Delta_{k-1}, \qquad \varphi(m+1) = \varphi(m) + \Delta_m,$$

one obtains

$$\varphi(m)\Delta_{k-1} + \varphi(k-1)\varphi(m+1) = \varphi(k)\varphi(m+1) - \Delta_{k-1}\Delta_m.$$

Therefore

$$F_m(\Delta_{k-1}t) = \frac{\alpha + t(\varphi(k)\varphi(m+1) - \Delta_{k-1}\Delta_m)}{\varphi(m+1)}.$$

It is enough to prove

$$M_\alpha(t) \geq t + \frac{\alpha - \Delta_{k-1}\Delta_m t}{\varphi(k)\varphi(m+1)}.$$

Apply the product lemma with $i = k-1$ and $j = m$. Then

$$\ell = (k-1+1)(m+1) - 1 = k(m+1) - 1,$$

and

$$B_\ell \leq B_{k-1}B_m = \varphi(k)\varphi(m+1),$$

while

$$\Delta_\ell \leq \Delta_{k-1}\Delta_m.$$

The affine term indexed by ℓ in $M_\alpha(t)$ gives

$$M_\alpha(t) \geq \frac{\varphi(\ell)t + \alpha}{\varphi(\ell+1)}.$$

Since

$$B_\ell = \varphi(\ell+1), \qquad \Delta_\ell = \varphi(\ell+1) - \varphi(\ell),$$

we have

$$\varphi(\ell) = B_\ell - \Delta_\ell.$$

Hence

$$M_\alpha(t) \geq t + \frac{\alpha - \Delta_\ell t}{B_\ell}.$$

If

$$\alpha - \Delta_{k-1}\Delta_m t < 0,$$

then

$$t + \frac{\alpha - \Delta_{k-1}\Delta_m t}{\varphi(k)\varphi(m+1)} < t,$$

and the desired inequality follows from $M_\alpha(t) \geq t$.

If

$$\alpha - \Delta_{k-1}\Delta_m t \geq 0,$$

then

$$\alpha - \Delta_\ell t \geq \alpha - \Delta_{k-1}\Delta_m t \geq 0,$$

and

$$B_\ell \leq \varphi(k)\varphi(m+1).$$

Therefore

$$\frac{\alpha - \Delta_\ell t}{B_\ell} \geq \frac{\alpha - \Delta_{k-1}\Delta_m t}{\varphi(k)\varphi(m+1)}.$$

Thus the affine-term case follows.

The estimate has now been checked for s, for α, and for every affine term indexed by m. Hence it holds for $M_\alpha(s)$. $\square$

6 A comparison lemma for the Schlumprecht norm

Define iterative norms on c_{00} by

$$\|x\|_0 = \|x\|_\infty,$$

and

$$\|x\|_{r+1} = \max\left\{\|x\|_\infty, \sup_{n\geq 2,\ E_1<\cdots<E_n} \frac{1}{\varphi(n)} \sum_{i=1}^{n} \|E_i x\|_r\right\}.$$

Then

$$\|x\|_{\mathcal{S}} = \sup_{r\geq 0} \|x\|_r.$$

Lemma (Local comparison)**.** *Let $y \in c_{00}$. Suppose there is a function*

$$\rho : [\mathbb{N}]^{<\omega} \to [0, \infty),$$

where

$$[\mathbb{N}]^{<\omega} = \{F \subset \mathbb{N} : F \text{ is finite}\},$$

such that, for every finite $F \subset \mathbb{N}$,

$$\|Fy\|_\infty \leq \rho(F),$$

for every $G \subset F$,

$$G \subset F \Longrightarrow \rho(G) \leq \rho(F),$$

and, whenever $k \geq 2$ and $E_1 < \cdots < E_k \subset F$,

$$\sum_{i=1}^{k} \rho(E_i) \leq \varphi(k)\rho(F).$$

Then

$$\|Fy\|_{\mathcal{S}} \leq \rho(F)$$

for every finite $F \subset \mathbb{N}$.

Proof. We prove by induction on r that

$$\|Fy\|_r \leq \rho(F)$$

for every finite F. The case $r = 0$ is exactly the assumption $\|Fy\|_\infty \leq \rho(F)$.

Assume the claim holds for r. For $E_1 < \cdots < E_k \subset F$,

$$\sum_{i=1}^{k} \|E_i y\|_r \leq \sum_{i=1}^{k} \rho(E_i) \leq \varphi(k)\rho(F).$$

Together with $\|Fy\|_\infty \leq \rho(F)$, this gives

$$\|Fy\|_{r+1} \leq \rho(F).$$

Taking the supremum over r proves the result. □

7 Far-coordinate perturbation

Proposition. *Let $0 < \alpha < 1$, let $x \in c_{00}$, and assume*

$$\|x\|_{\mathcal{S}} < a_\alpha.$$

If $N > \max \operatorname{supp} x$, then

$$\|x + \alpha e_N\|_{\mathcal{S}} \le 1 \qquad \text{and} \qquad \|x - \alpha e_N\|_{\mathcal{S}} \le 1.$$

Proof. By 1-unconditionality, it is enough to prove the estimate for $|x| + \alpha e_N$. Thus assume $x \ge 0$, and set

$$y = x + \alpha e_N.$$

For each finite $F \subset \mathbb{N}$, define

$$\rho(F) = \begin{cases} \|Fx\|_{\mathcal{S}}, & N \notin F, \\ M_\alpha(\|Fx\|_{\mathcal{S}}), & N \in F. \end{cases}$$

The lattice monotonicity of the Schlumprecht norm and the monotonicity of M_α give

$$G \subset F \Longrightarrow \rho(G) \le \rho(F).$$

Also,

$$\|Fy\|_\infty \le \rho(F),$$

because $M_\alpha(u) \ge u$ and $M_\alpha(u) \ge \alpha$.

It remains to verify the local comparison inequality. Fix finite $F \subset \mathbb{N}$, and let

$$E_1 < \cdots < E_k \subset F, \qquad k \ge 2.$$

If $N \notin F$, then

$$\sum_{i=1}^k \rho(E_i) = \sum_{i=1}^k \|E_i x\|_{\mathcal{S}} \le \varphi(k)\|Fx\|_{\mathcal{S}} = \varphi(k)\rho(F).$$

Assume now that $N \in F$. If none of the E_i contains N, then

$$\sum_{i=1}^k \rho(E_i) = \sum_{i=1}^k \|E_i x\|_{\mathcal{S}} \le \varphi(k)\|Fx\|_{\mathcal{S}} \le \varphi(k)M_\alpha(\|Fx\|_{\mathcal{S}}) = \varphi(k)\rho(F).$$

Finally suppose exactly one set E_j contains N, and fix this j. Then

$$\sum_{i=1}^k \rho(E_i) = M_\alpha(\|E_j x\|_{\mathcal{S}}) + \sum_{i \ne j} \|E_i x\|_{\mathcal{S}}.$$

The remaining pieces satisfy two estimates. First,

$$\sum_{i \ne j} \|E_i x\|_{\mathcal{S}} \le \varphi(k-1)\|Fx\|_{\mathcal{S}}.$$

For $k = 2$, this is just monotonicity, since $\varphi(1) = 1$. For $k > 2$, it follows from the Schlumprecht recursive inequality applied to the $k-1$ remaining blocks. Second, using the full k-tuple,

$$\|E_j x\|_{\mathcal{S}} + \sum_{i \ne j} \|E_i x\|_{\mathcal{S}} \le \varphi(k)\|Fx\|_{\mathcal{S}},$$

so

$$\sum_{i\neq j}\|E_ix\|_{\mathcal S}\le\varphi(k)\|Fx\|_{\mathcal S}-\|E_jx\|_{\mathcal S}.$$

Therefore

$$\sum_{i\neq j}\|E_ix\|_{\mathcal S}\le\min\{\varphi(k-1)\|Fx\|_{\mathcal S},\varphi(k)\|Fx\|_{\mathcal S}-\|E_jx\|_{\mathcal S}\}.$$

By the strengthened scalar estimate,

$$\begin{aligned}\sum_{i=1}^{k}\rho(E_i)&\le M_\alpha(\|E_jx\|_{\mathcal S})+\min\{\varphi(k-1)\|Fx\|_{\mathcal S},\varphi(k)\|Fx\|_{\mathcal S}-\|E_jx\|_{\mathcal S}\}\\&\le\varphi(k)M_\alpha(\|Fx\|_{\mathcal S})=\varphi(k)\rho(F).\end{aligned}$$

The local comparison lemma gives

$$\|Fy\|_{\mathcal S}\le\rho(F)$$

for every finite F. Taking $F=\operatorname{supp}x\cup\{N\}$, we get

$$\|x+\alpha e_N\|_{\mathcal S}\le M_\alpha(\|x\|_{\mathcal S})\le 1,$$

because $\|x\|_{\mathcal S}<a_\alpha$. The estimate for $x-\alpha e_N$ follows by unconditionality. □

8 The reverse Szlenk-radius inclusion

Let $0<\varepsilon<2$, and put

$$\beta=\varepsilon/2.$$

We prove

$$r_{\mathcal S^*}(\varepsilon)\ge a_\beta.$$

First take $x\in c_{00}$ with

$$\|x\|_{\mathcal S}<a_\beta.$$

Choose α such that

$$\beta<\alpha<1\qquad\text{and}\qquad\|x\|_{\mathcal S}<a_\alpha.$$

This is possible because the map $\alpha\mapsto a_\alpha$ is continuous: it is the infimum of affine functions

$$\alpha\mapsto\frac{\varphi(n+1)-\alpha}{\varphi(n)}$$

whose slopes have absolute value at most 1.

For every sufficiently large N, the far-coordinate perturbation result gives

$$x+\alpha e_N\in B_{\mathcal S},\qquad x-\alpha e_N\in B_{\mathcal S}.$$

Since $e_N\to 0$ weakly, both points converge weakly to x. Their distance is

$$\|(x+\alpha e_N)-(x-\alpha e_N)\|_{\mathcal S}=\|2\alpha e_N\|_{\mathcal S}=2\alpha>\varepsilon.$$

Therefore every weak neighbourhood of x meets $B_{\mathcal S}$ in a set of diameter $>\varepsilon$. Hence

$$x\in s_\varepsilon(B_{\mathcal S}).$$

Now let $r<a_\beta$. If $z\in rB_{\mathcal S}$, choose finitely supported $x_n\to z$ in norm with $\|x_n\|_{\mathcal S}<a_\beta$ for all large n. By the finite-support case,

$$x_n\in s_\varepsilon(B_{\mathcal S})$$

eventually. The set $s_\varepsilon(B_{\mathcal{S}})$ is weakly closed, hence norm closed, so

$$z \in s_\varepsilon(B_{\mathcal{S}}).$$

Thus

$$rB_{\mathcal{S}} \subset s_\varepsilon(B_{\mathcal{S}}) \qquad (r < a_\beta),$$

and therefore

$$r_{\mathcal{S}^*}(\varepsilon) \geq a_\beta.$$

Together with the known upper bound, this gives

$$r_{\mathcal{S}^*}(\varepsilon) = a_{\varepsilon/2} = \inf_{n\in\mathbb{N}} \frac{\varphi(n+1) - \varepsilon/2}{\varphi(n)}.$$

Comment on notation

The original proof introduced several temporary letters for quantities such as $\varphi(k)$, $\varphi(m+1)$, Δ_{k-1}, Δ_m, B_ℓ, and Δ_ℓ. This rewrite keeps the logarithmic weights and increments visible. In particular, the key use of the product lemma in the scalar estimate is simply

$$B_\ell \leq \varphi(k)\varphi(m+1), \qquad \Delta_\ell \leq \Delta_{k-1}\Delta_m,$$

where

$$\ell = k(m+1) - 1.$$

Human Review: Schlumprecht Szlenk-radius converse

Original automatic proof: `solution_packet.pdf`
Companion notation rewrite: `human_requested_better_notation.pdf`
Checked date: 14 June 2026
Status: Manually checked.
Verdict: Appears correct.

Human Comments

The proposed proof appears to be correct. It gives the reverse inequality for the Szlenk-radius formula in Schlumprecht's space, and therefore appears to complete the converse left open in the source paper.

The idea of the proof is to show that, if a vector has norm smaller than the proposed radius, then, after harmlessly reducing to the finitely supported case, one can perturb it far out by

$$\pm\alpha e_N,$$

where $\alpha > \varepsilon/2$. These two perturbations are

$$x + \alpha e_N \quad \text{and} \quad x - \alpha e_N.$$

They are weakly close to x, because $e_N \to 0$ weakly, but they are separated by

$$\|(x + \alpha e_N) - (x - \alpha e_N)\|_{\mathcal{S}} = 2\alpha > \varepsilon.$$

Thus they witness diameter $> \varepsilon$ in every weak neighbourhood of x, provided they remain in the unit ball.

The key technical step is exactly this unit-ball control:

$$\|x \pm \alpha e_N\|_{\mathcal{S}} \leq 1$$

whenever

$$\|x\|_{\mathcal{S}} < a_\alpha, \qquad a_\alpha = \inf_{n\geq 1} \frac{\varphi(n+1) - \alpha}{\varphi(n)}.$$

This is the part where one has to control the size of the new vectors in the recursive Schlumprecht norm. The scalar majorant M_α, the logarithmic product lemma, and the local comparison argument are all used to prove precisely this point.

I checked the main scalar estimate and the comparison argument. The argument appears sound. The original notation was somewhat difficult to follow, so I requested a separate improved-notation rewrite of the proof. That rewrite is included only as a human-requested notational aid; it is not the original automatic output.

Short Version

The proof works by taking x below the proposed radius and perturbing it far out by $\pm\alpha e_N$. The perturbations are weakly close to x and are more than ε apart. The main work is to show that both perturbed vectors remain in the unit ball of Schlumprecht's space. Once this is proved, the reverse Szlenk-radius inclusion follows by the standard density and weak-closedness argument.

Literature and Follow-Up

This appears to be a serious candidate full solution to the converse problem in the source paper. I recommend checking the argument with the authors, both to confirm correctness and to understand whether the result or method was already known to them.

Review Outcome

Archive this packet as an apparently correct candidate full solution. The main mathematical content is the far-coordinate perturbation argument and the scalar estimate controlling the Schlumprecht norm after adding $\pm\alpha e_N$.